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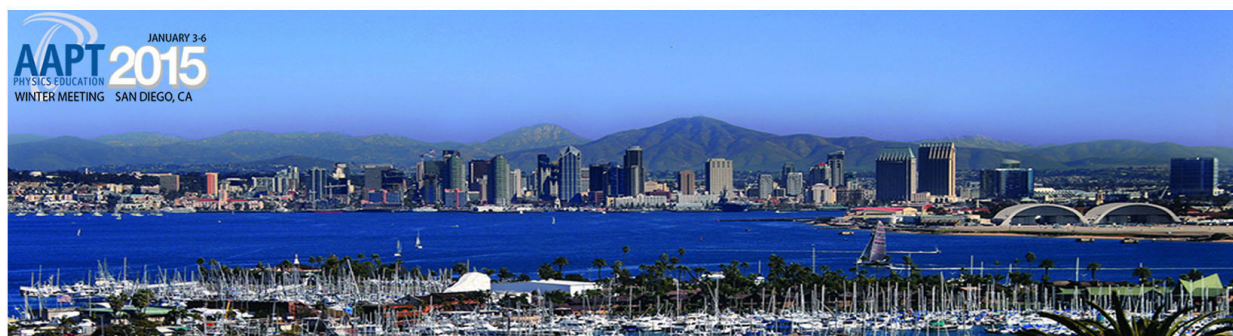
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A Simple Treatment of Potential Barrier Penetration*

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It is pointed out that the penetration of a particle through a potential barrier may be understood as a climb over the barrier during one of the energy fluctuations expected from the uncertainty principle, rather than as a "tunneling" under the barrier. A method of obtaining the quantitative result with this model is presented.

ONE of the most difficult tasks in expounding on atomic or nuclear phenomena to nonphysicists is to explain the phenomenon of potential barrier penetration. The most popular explanation is in terms of "tunneling," a concept which is completely contrary to whatever knowledge of physics students may have. The idea that a system can have a negative kinetic energy—i.e., an imaginary velocity—is most unappealing to such an audience.

It is the purpose of this note to suggest a more satisfying physical explanation, which, with the aid of a simple, intuitively appealing generalization and sophomore-level mathematics, even gives an accurate quantitative result. This explanation is based on the uncertainty principle¹

$$\Delta E \cdot \Delta t \simeq \hbar, \quad (1)$$

which is familiar to millions of nonscientists (including high-school students) and which, in some of its facets, can be demonstrated experimentally. Eq. (1) may be interpreted as meaning that the energy of a system is not constant, but subject to fluctuations for short times consistent with (1). This interpretation is frequently used in explaining the range of the nuclear force in terms of the meson mass. For the present case, it introduces the possibility that a particle confined by a potential barrier, can get through if its energy fluctuates sufficiently to keep it above the barrier height long enough for it to cross the length of the barrier. It is this possibility that we treat in a semiquantitative way.

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¹ The uncertainty principle $\Delta p \cdot \Delta x \simeq \hbar$ is used by R. Gomer, in *Field Emission and Field Ionization* (Harvard University Press, Cambridge, Massachusetts, 1961) to show when barrier penetration is large, but this treatment is not useful in situations where barrier penetration is small as in alpha decay.

Since ΔE in (1) means the total fluctuation (on both sides of E), one can expect to find the energy larger than E by an amount ϵ for a time interval τ given by

$$\epsilon \simeq \frac{1}{2} (\hbar / \tau). \quad (2)$$

Of course, one can also expect the energy to fluctuate to values smaller than E by this same amount, but this does not interest us here. It seems intuitively reasonable to put this on a more quantitative basis by saying that the probability P , for a fluctuation ϵ for a time τ is

$$P \sim \exp(-2\epsilon\tau/\hbar). \quad (3)$$

As in other applications of the uncertainty principles, (3) is certainly not rigorously correct. It also has the problem of not being normalizable in two dimensions without cutoffs, although there are good physical reasons for cutoffs in our application. (Actually, the usual quantum mechanical treatments of barrier penetration given in graduate physics courses also have normalization difficulties.) However, (3) has an intuitive appeal that makes it easily acceptable to people seeking only a simple physical understanding.

Once (3) is accepted, the treatment of the problem may be geared to the mathematical sophistication of the audience; we give here the most sophisticated treatment that seems reasonable.

Let us say the potential is $V(x)$, and the region where V is greater than E extends from x_0 to x_n . The kinetic energy is $E + \epsilon - V$, so that the velocity v is

$$v = \left[\frac{2}{m} (E + \epsilon - V) \right]^{\frac{1}{2}}, \quad (4)$$

where v and ϵ may be functions of x . The prob-

ability for the particle to travel the infinitesimal distance dx from x_0 to x_1 is, from (3),

$$P_1 \simeq \exp(- (2\epsilon/\hbar)(dx/v)).$$

The probability P , for the particle to get from x_1 to x_n is just the product of the probabilities for getting from x_0 to x_1 , from x_1 to x_2 , $\dots$ from x_{n-1} to x_n . Thus

$$P \simeq \prod_{i=1}^n \exp\left(-\frac{2\epsilon_i}{\hbar} \frac{dx_i}{v_i}\right) = \exp\left(-\sum_i \frac{2\epsilon_i}{\hbar} \frac{dx_i}{v_i}\right),$$

or in the limit as dx_i goes to zero,

$$P \simeq \exp\left(-\int_{x_0}^{x_n} \frac{2\epsilon}{\hbar} \frac{dx}{v}\right). \quad (5)$$

Inserting (4) into (5), the latter can be written

$$P \simeq \exp\left(- (2m)^{\frac{1}{2}} \frac{I}{\hbar}\right), \quad (6)$$

where

$$I = \int_{x_0}^{x_n} \epsilon(\epsilon - V + E)^{-\frac{1}{2}} dx. \quad (7)$$

We have still not specified the function $\epsilon(x)$. It seems reasonable to choose this so as to minimize I , and thus to maximize the probability. (In any case, the result is quite insensitive to the choice of $\epsilon(x)$ so long as it is of the order of the barrier height.) Thus we set the partial derivative of the integrand in (7) with respect to ϵ equal to zero. This gives

$$(\epsilon - V + E)^{-\frac{1}{2}} - \frac{1}{2} \epsilon (\epsilon - V + E)^{-\frac{3}{2}} = 0,$$

whence

$$\begin{aligned} \epsilon - V + E - \frac{1}{2} \epsilon &= 0, \\ \epsilon &= 2(V - E). \end{aligned} \quad (8)$$

Inserting (8) into (7),

$$I = 2 \int_{x_0}^{x_n} (V - E)^{\frac{1}{2}} dx, \quad (9)$$

which may then be substituted into (6) to give

$$P \simeq \exp\left(-2 \int_{x_0}^{x_n} \left[\frac{2m}{\hbar^2} (V - E)\right]^{\frac{1}{2}} dx\right). \quad (10)$$

This is identical with the accurate result as derived quantum mechanically by use of the WKB method. This agreement should not be taken too seriously because of the questionable nature of the basic assumption, Eq. (3), but it is certainly a matter of great convenience in such applications as alpha decay where numerical results are highly sensitive to small changes in the formula.

Where less sophisticated mathematics is called for, the minimization can be eliminated and any simple function $\epsilon(x)$ can be used. Actually, the minimum is a very broad one; for example, if $V(x)$ is linear and ϵ is taken as a constant of any reasonable value [quite a deviation from (8)], the integral I is changed by only about 20%. Where an even simpler treatment is desired, the integral can be replaced by "average" values; by a judicious choice of "averages," the accurate result is readily obtained. In any case, all the functional dependences are obtained without difficulty even for a high-school-level audience.

But perhaps the most important advantage of this approach is that it eliminates the disturbing idea of "tunneling," and replaces it with a widely understood concept that is useful in many other connections. While no physical explanation can be called "correct" or "incorrect," the "tunneling" concept is certainly not intuitively appealing. It seems preferable not to say that the particle tunnels under the barrier; we may say instead that it climbs over the barrier during one of the fluctuations of its energy.

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